جامعة خليفة
Khalifa University

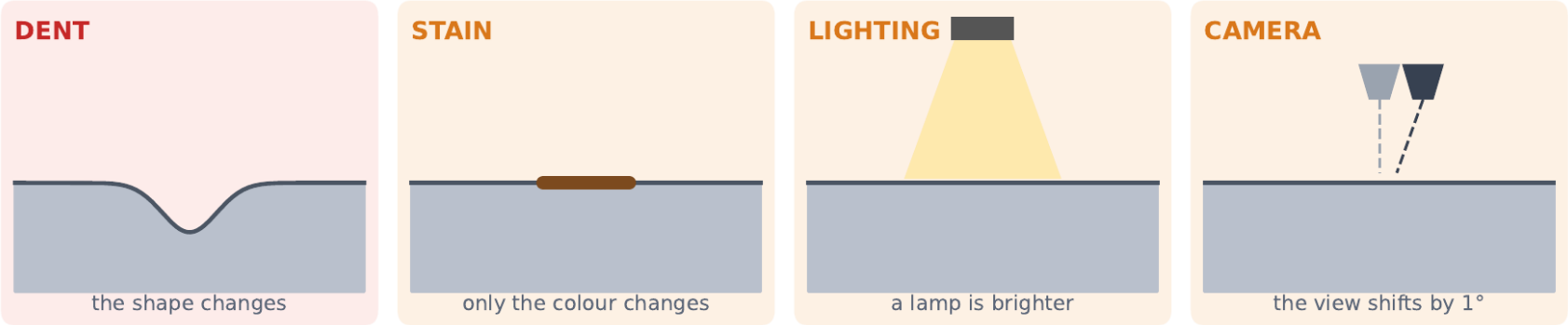
innovation oasis
واحة الابتكار
Part of Silal جزء من سلال

When Do Images Prove a Defect?

Evidence-bounded 3D defect estimation for aero-engine blades

AFEEFA AZAM · KHALIFA UNIVERSITY

A dark region in an image has several physically different causes



- **Signal scale.** A 0.1 mm dent changes a few hundred pixels by a few grey levels; a 1° pose error or a 10% lighting change alters thousands.
- **Consequence.** A detector that flags any unexplained change confuses appearance with geometry; one that must always answer reports damage the images do not support.
- **Requirement.** Separate geometric change from pose, illumination and reflectance, and abstain when they cannot be separated.

We pose defect estimation as a calibrated test that can abstain

	Definition
Input	Healthy CAD mesh + K = 2–7 calibrated views
H0 (nuisance)	Per-view pose, light gains, smooth brightness field, local albedo patch
H1 (damage)	H0 + a continuous parametric surface displacement
Evidence	gain = L(H0) − L(H1), robust image loss
Decision	Report if gain > τ, else abstain
Calibration	τ = split-conformal quantile on healthy + stained blades (α = 5%)

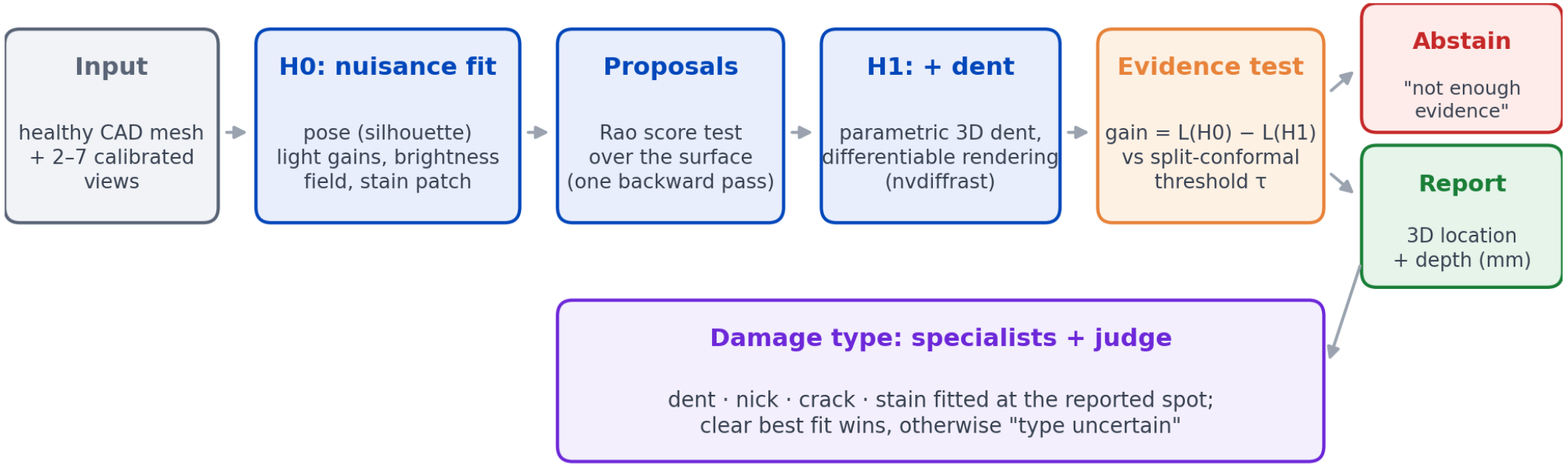
- **Constraints.** Calibrated rig, fixed light positions, appearance change only through the brightness field and a local patch. Without such constraints, nuisance can explain any change.
- **Output.** A per-vertex displacement (location, depth) when supported; otherwise “not supported by these views”.
- **Guarantee.** False-change rate controlled on negatives exchangeable with the calibration set.

Prior work either localises anomalies in 2D or reconstructs a twin; none abstains

Line of work	Representative	Output	3D change (mm)	Abstains
Pose/illumination-agnostic AD	PAD (NeurIPS 23), SplatPose (24), PIAD (CVPR 25)	2D anomaly map	✗	✗
Reference/CAD-based AD	Looking 3D (CVPR 24)	Class, box, mask	✗	✗
CAD-prior reconstruction	CAD-NeRF (24), CADSplat (26)	Digital twin	✗	✗
Blade defect datasets	BladeSynth (Sci. Data 25)	2D detection	✗	✗
3D AD benchmarks	MVTec 3D-AD, Real3D-AD, SiM3D	Anomaly score	✗	✗
This work	**Render-and-test with calibration**	**3D displacement or abstain**	✓	✓

- Building blocks: differentiable rasterisation (nvdiffrast), score and likelihood-ratio tests, split-conformal calibration.

Method: nuisance profiling, score-test proposals and a calibrated decision



- **Design choices.** pose from silhouettes (lighting-independent); evidence on interior, non-grazing pixels; dent and stain searched symmetrically so a stain cannot hide a dent.

Worked example: the evidence exceeds the threshold by a wide margin

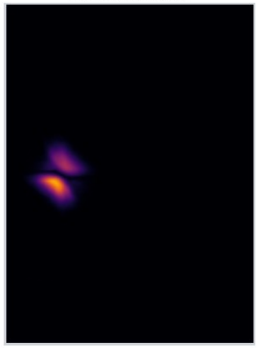
Photo



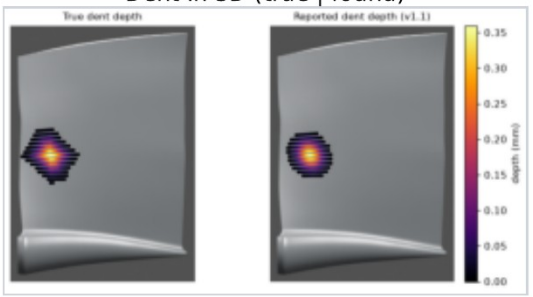
Healthy drawing



Difference



Dent in 3D (true | found)

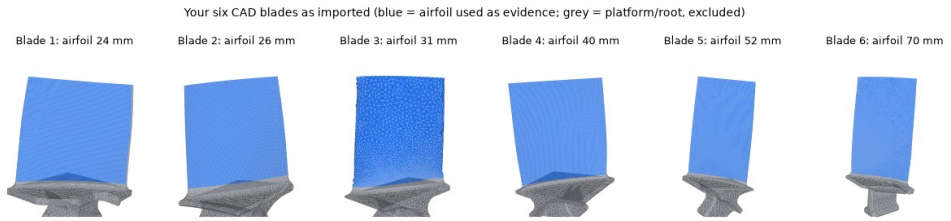


Quantity	Value
L(H0), no damage	89.3
L(H1), with dent	19.9
Evidence (gain)	**69.4**
Threshold τ	1.02
Depth found / true	0.35 / 0.36 mm

Sealed test case test2_0083_3 (Rotor37, unseen blade).

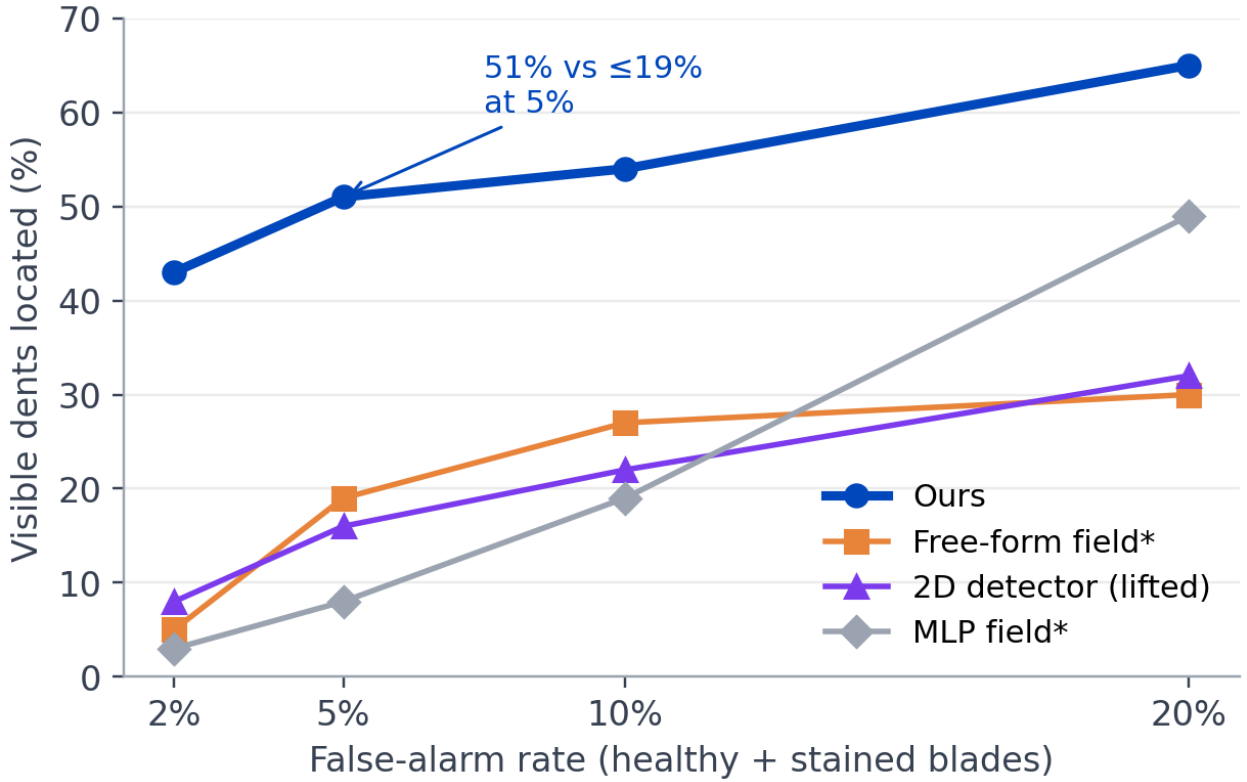
BladeDelta: 1,352 rendered cases with exact ground truth

Split	Cases	Role
Rotor37 dev / calib	96 / 120 (+300)	Development, calibration
Rotor37 sealed test	240 (30 blades)	Main result
Rotor37 second test	240 (30 blades)	Reproduction
Six industrial designs	456	Transfer (blades 3–6 held out)
Defect types	136 + 64	Nicks, cracks, stains



- Blender (Cycles) renders; damage applied to the mesh
- Controlled camera error (≤1.5°) and lighting change (±15%)
- Damage families differ from the estimator’s model
- Test splits sealed until the method was frozen

At equal false-alarm rates, the method locates 2.7–6× more dents

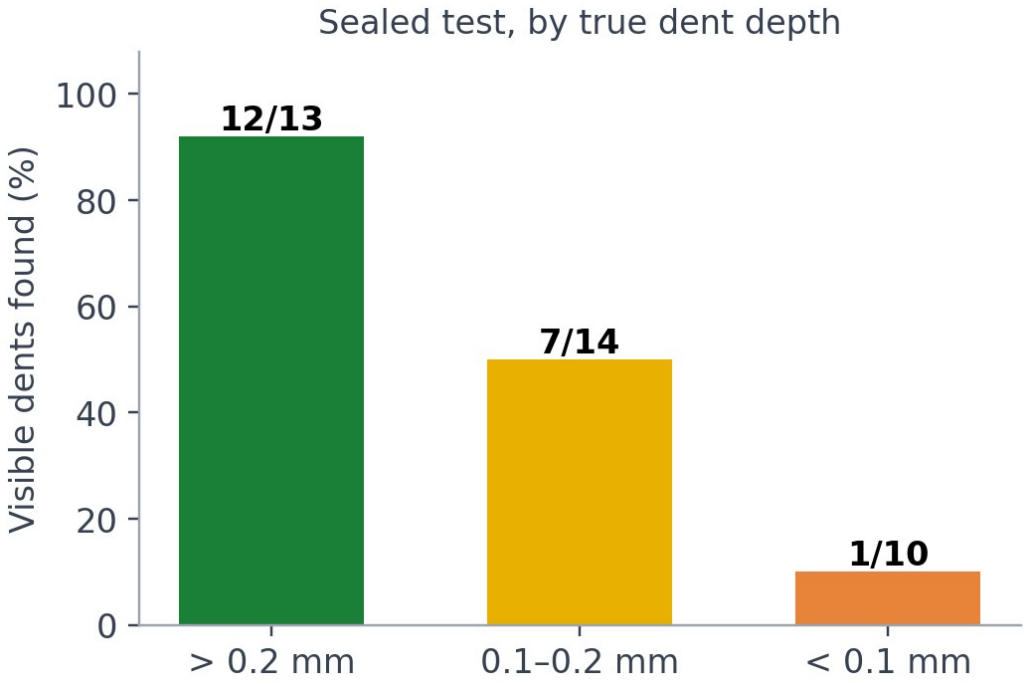


Method	AUROC	Located @5%
Ours	**0.82**	**51%**
Free-form field*	0.63	19%
2D detector (lifted)	0.72	16%
MLP field*	0.74	8%

- Second sealed test reproduces it (46% located, AUROC 0.79)
- 12 of 13 visible dents deeper than 0.2 mm found; median centre error 0.7 mm

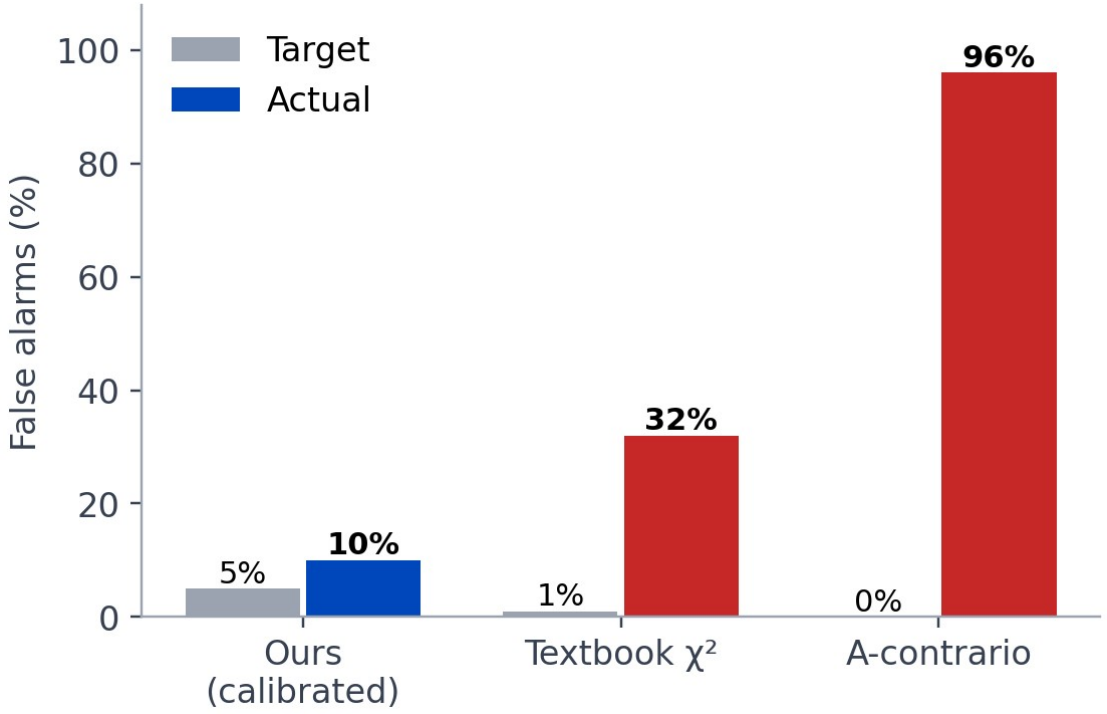
*Sealed test, 240 cases, 30 unseen Rotor37 blades; 90 negatives, 37 visible dents. *Re-implementations of the CADSplat idea.*

Misses concentrate below 0.2 mm, where the image signal is weakest



- **Detectability is predictable.** The profiled information of a dent predicts found vs missed with AUROC 0.90 / 0.95 on the two sealed tests.
- **Safety-weighted recall.** Weighting misses by depth: 73%; critical defects (> 0.2 mm): 92%.
- **Per-point maps are optimistic.** They imply 49–66% recall below 0.1 mm against 10% observed; reported as an upper bound.

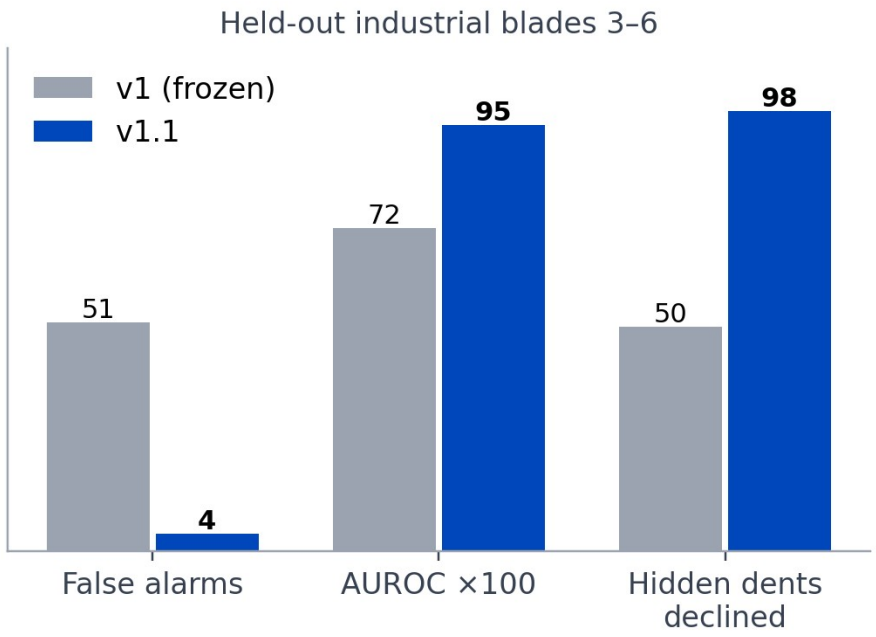
Calibration is necessary: textbook thresholds do not control false alarms



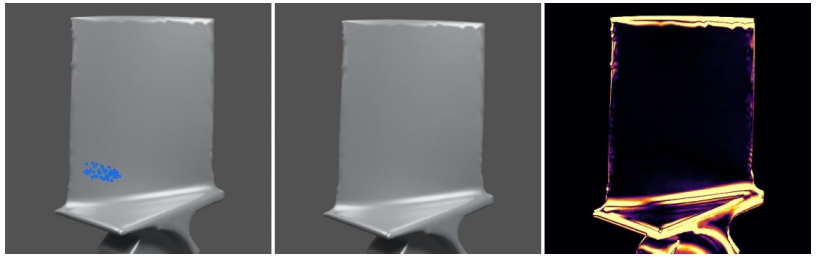
- **Why.** Textbook thresholds assume ideal pixel noise; the residual after the best nuisance fit is dominated by model mismatch.
- **χ^2.** Nominal 1% → 32% false alarms (≈30×).
- **A-contrario.** Flags 96% of healthy blades.
- **Ours.** Recalibrated on 360 negatives: 2% target → 4% observed.

Same fits on the sealed test.

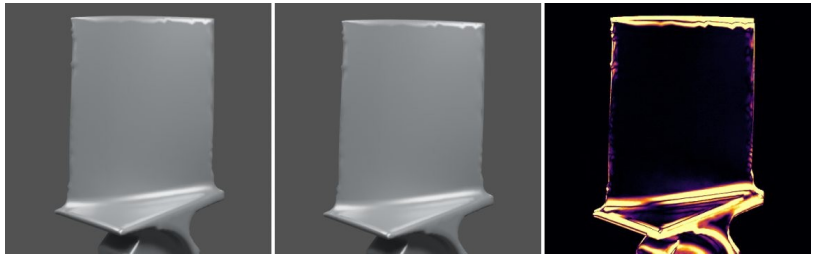
Two corrections restore transfer to unseen industrial designs



v1 (frozen): a false dent on a healthy blade



v1.1: correctly nothing reported

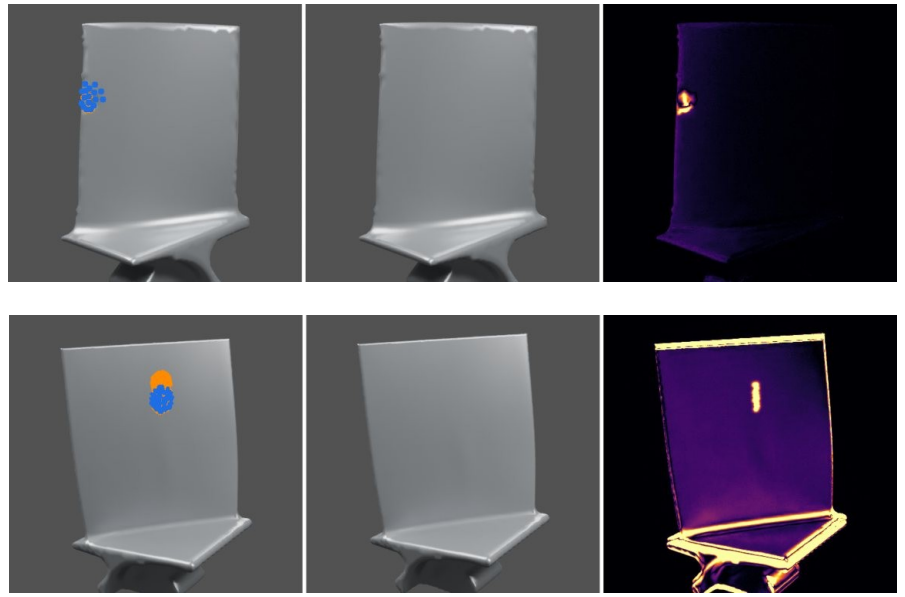


- **Causes.** a silhouette pose bias, and a healthy-reference correction that did not move with the camera. Fixed on blades 1–2, no retraining; evaluated on 3–6.

Detection extends to nicks and cracks without retraining

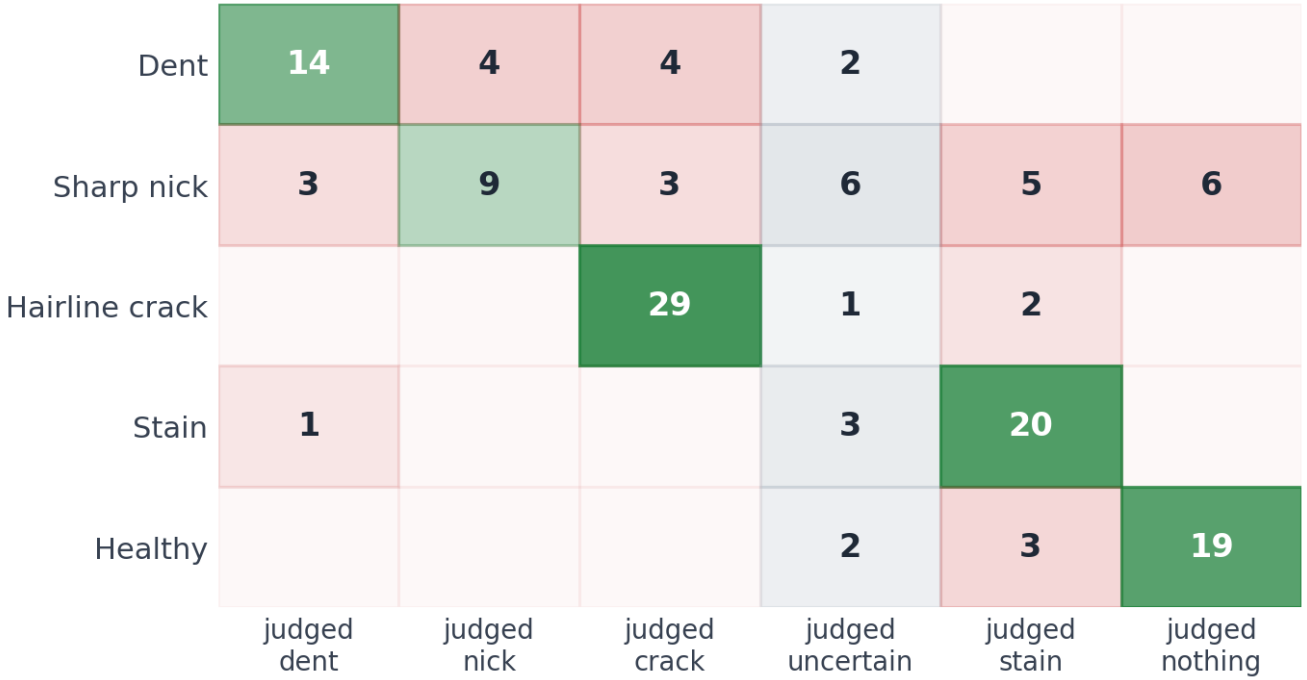
Defect (held-out designs)	Cases	Ours: located	SIFT: reported
Dent	24	23 (96%)	6 (25%)
Sharp nick (edge notch)	32	21 (66%)	9 (28%)
Hairline crack (dark line)	32	22 (69%)	15 (47%)
Stain (should be 0)	24	4 reported	3
Healthy (should be 0)	24	2 reported	3

- False alarms 6/48 (12.5%) exceed the 5% target set on Rotor37: recalibration on these designs is required.



Sharp nick (top), hairline crack (bottom): orange = truth, blue = found

Physical specialists and a judge recognise the damage type



22 / 22
cracks typed correctly

9 / 15
nicks typed; 6 more uncertain

42 / 43
location preserved

2 → 9
nicks recognised, round 1 → 2

- Specialists: smooth dent, soft-ellipsoid notch, dark-line crack, colour patch
- Judge: lowest loss wins; “uncertain” if within τ. Adapted from MedGuards

Nick and crack: 32 realistic cases each (round 2); dent, stain, healthy: 24 each. Stain/nothing on damage = detector miss.

Location is identifiable from two views; depth is not



Unseen design: correct location, depth 0.06 mm for a true 0.64 mm

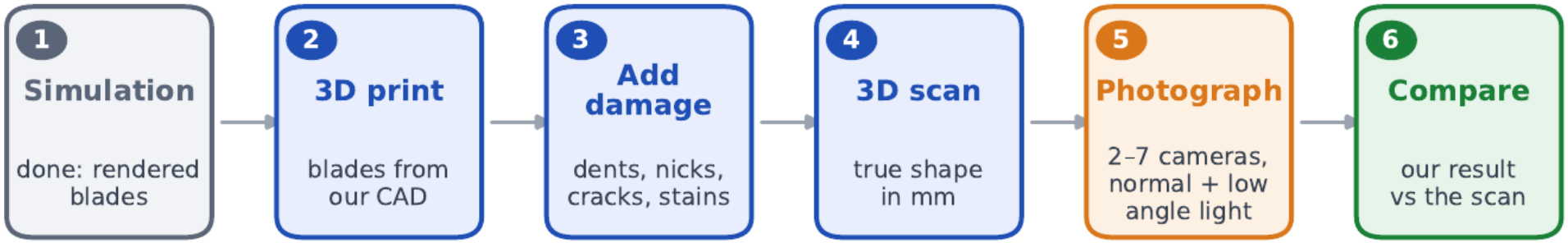
Depth, measured / true (median)	
Rotor37 sealed test	0.89
Industrial held-out	0.60
4 views instead of 2	no change (0.68 vs 0.71)

- **Not a search failure.** Better proposals and restarts at the true location recover ≤ 4 of 42 misses.
- **Not ambiguity alone.** Profile-likelihood depth intervals are narrow yet contain the truth in 36 of 109 cases: a systematic bias.
- **Implication.** Report location; treat depth as unresolved under passive two-view capture. Controlled (grazing) illumination is the next lever.

Limitations

- **Rendered data only.** All positive results use rendered images with known materials.
- **Real scans fail so far.** On SiM3D real objects: AUROC 0.54, 0/22 defects detected; dominated by misregistration and texture.
- **Depth bias on new designs.** Median 60% of true depth; not fixed by views or intervals.
- **Calibration transfer.** 12.5% false alarms on the defect-type set vs 5% target; requires per-design recalibration.
- **Nicks.** 11 of 32 not detected; 6 of 21 located nicks mistyped.
- **Realism gaps.** No sensor noise, blur or material variation in the renders yet.

Next: physical validation with measured ground truth



- **Specimens.** 10 stainless-steel printed blades; dents (ball bearings), nicks (files), stains (paint).
- **Ground truth.** Dial gauge and 3D scanner.
- **Capture.** 2–7 calibrated cameras, ArUco markers, fixed and grazing light.
- **Method.** Per-design and class-specific thresholds; grazing-light depth.
- **Baselines.** PatchCore; VGGT + Sim(3) alignment.
- **Realism.** Sensor noise, blur, material variation in the benchmark.

Summary of contributions

- **Formulation.** CAD-referenced defect estimation as a calibrated test with abstention, under explicit capture constraints.
- **Method.** Nuisance profiling, score-test proposals and differentiable rendering; 51% vs ≤ 19% at equal false alarms.
- **Calibration.** Shown to be necessary: textbook tests give 32–96% false alarms.
- **Identifiability finding.** Location transfers across designs; depth does not from two passive views.
- **Defect typing.** Physical specialists and a judge: cracks 22/22, nicks 9/15 with explicit uncertainty.
- **Benchmark.** 1,352 rendered cases on public and industrial aero-engine blades.

جامعة خليفة
Khalifa University

Thank You

Dashboard: afeefaaazam03.github.io/3D-Jetblade/visual_results.html